ANUSHESH JUMALE | 23IE10005

B.Tech.(Hons.) in INSTRUMENTATION ENGINEERING

EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	B.TECH	IIT Kharagpur	6.79 / 10
2022	CBSE XII	Jawahar Navodaya Vidyalaya	90.4%
2020	CBSE X	Jawahar Navodaya Vidyalaya	91.6%

INTERNSHIPS

Computer Vision Intern | Dewinter Optical Inc. | New Delhi **[May'26 - Jul'26]**

(Transformed the legacy metallurgical and biological analysis softwares into a fully AI-driven system)

- Deployed **U-Net, Cascade Mask R-CNN**, and **YOLOv** models in PyTorch, achieving **92% mAP** for biological and metallurgical microscopy analysis
- Engineered **SAHI** pipelines using **512x512** sliding-window slicing and **TTA**, boosting mask boundary accuracy by **14%** on gigapixel size images
- Built offline **PySide6** microscopy annotator integrating **SAM**, **Cellpose**, and **classical CV** pipelines for automated cell/nuclei segmentation
- Built multithreaded Qt architecture for async **ML** inference, custom undo/redo engine and **COCO/YOLO/mask** exporters for **ML** pipelines

Software Engineering Intern | Zip Softwares |Remote **[Jun'25 - Jul'25]**

(Developed core modules for a comprehensive Hospital Management System)

- Developed a **React** and **PostgreSQL** dashboard for real-time tracking of **300+** daily admissions, boosting overall hospital resource efficiency
- Engineered **Chart.js** and **Python** data visualizations, replacing manual reporting and accelerating key workflows and revenue insights by **30%**
- Integrated Razorpay API for secure, automated online billing of about **8,000+** monthly consultations, reducing manual invoicing time by **45%**
- Implemented webhook listeners to synchronize payment statuses, processing **4,000+** transactions and eliminating **15+** reconciliation hours

PROJECTS

Spotify-Style Two-Stage Recommender System **[Mar'26 - Apr'26]**

- Architected scalable recommendation model for **1.15M** songs, deploying **FastAPI** server to instantly serve **top-10** personalized result in **120ms**
- Engineered a PyTorch Two-Tower retrieval model, using **FAISS** vector searches to instantly filter **1.15M** tracks down to **100** prime candidates
- Developed an **XGBoost** precision ranker on **250K** interactions using **4:1** negative sampling, achieving a **0.81 ROC-AUC** score and **80%** recall
- Built a **Parquet** data pipeline processing raw CSV/JSON sources, enforcing strict schema validation via **Pandera** to prevent train-serve skew

Legal Contract Analyzer **[May'26 - Jun'26]**

- Developed a full-stack **RAG** pipeline using a **27B LLM** (**Groq**) to power sub-second semantic retrieval and multi-document legal synthesis
- Processed **510+ CUAD** PDF contracts via PyMuPDF block extraction, embedding **2,400+** semantic text chunks into local **ChromaDB** vector store
- Mitigated LLM hallucinations using a strict constraint prompt, forcing the generation engine to append exact page citations to every claim
- Optimized cross-document context using context-aware splitting (**800**-char windows) and **top-10** dense search, achieving **<800ms** API latency

Distributed URL Shortener & Rate Limiter **[Jun'26 - Jul'26]**

- Architected a distributed URL shortener with sharded **Redis** caching via a custom **Consistent Hash Ring**, also cutting remap to **~1/N** on scaling
- Designed a **Snowflake**-style **64-bit** distributed ID generator (**41-bit** timestamp, **10-bit** worker, **12-bit** sequence) with global clock-drift handling
- Implemented an atomic **Lua**-scripted **Token Bucket** rate limiter in **Redis**, achieving **300+ req/s** with **100%** enforcement accuracy under load
- Built a fault-tolerant, highly resilient cache-aside system with a **3-state Circuit Breaker** (**PostgreSQL** failover) and async click-analytics pipeline

SKILLS AND EXPERTISE

Programming Languages: C | C++ | HTML | CSS | JavaScript | Python | SQL **Libraries/Frameworks:** React | PostgreSQL | Redis | Pytorch | Keras | TensorFlow | Pandas | NumPy | Matplotlib | Seaborn | Scikit-Learn | FastAPI | Transformers | YOLO | Langchain | Langgraph | FAISS
Softwares/Tools: MS Office Suite| Git | GitHub | VS Code | Jupyter Notebooks | Google Colab | Docker | AWS | Roboflow | Postman

POSITIONS OF RESPONSIBILITY

Finance Head | National Students' Space Challenge'25 **[Aug'23 - Apr'26]**

- Orchestrated end-to-end logistics with securing all administrative permissions and conceptualizing the arena design for **1,000+** attendees
- Managed a fiscal portfolio of INR **20 Lakhs**, successfully negotiating **20+** corporate associations to drive a **30%** YoY increase in net profits
- Led a workforce of **50+** members to execute **12** events and **5** guest lectures, ensuring seamless operational management and execution

Senior Executive Member | Space Technology Students' Society **[Jan'24- Present]**

- Published a **literature review** paper based on an analysis of **80+** research papers, achieving an H-index of **1** and over **3,600** full-text reads
- Organized Space Day for **400+** attendees, overseeing event & Exhibition logistics, team coordination, and end-to-end on-ground execution
- Successfully executed Space Technology Awareness Camps (STACs) across **8** universities, strategic planning and managing ground logistics

COURSEWORK INFORMATION

Probability and Statistics | Programming and Data Structures | Linear Algebra | Advanced Calculus | Economics | Applications of Machine Learning in Biological Systems | Digital Electronics | Signals and Systems | Digital Signal Processing | Transform Calculus | Control Systems

EXTRA CURRICULAR ACTIVITIES

- Engineered a cost-effective **SAW** sensing network in Inter-Hall Tech General Championship (**RailRaksha**), addressing railway tracks safety
- Secured **1st** place in the **Inter-Hall General Championship Hockey** tournament, representing Rajendra Prasad Hall at the middle position
- Invigilated the **Dakshana Selection Test** and conducted **mentorship sessions** to encourage students to strive for future academic success

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